

# QUANTUM JACKPOT

## Game Design Document

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# 1. Game Overview

Game Title - Quantum Jackpot | Format - 3 reels x 3 rows | 5 fixed paylines  
Theme - Sci - Fi Quantum Reactor | High Volatility | Feature - Reactor Overload

## 1.1. Concept

Quantum jackpot is set inside a super futuristic quantum computing facility. The visual centrepiece is the golden dilution refrigerator. Which looks like a chandelier with its gold rings and suspended superconductor cables. The player builds charge in the reactor overload meter by landing Energy Cells scatters. When it fills, the reactor fires and awards 8 free spins with a random Quantum Core multiplier active on every spin.

## 1.2. Player Loop

Base Game - Match symbol across 5 paylines for immediate win.  
Progression - Each Energy Cell in view adds one charge to the Reactor Overload Meter, building visible progress.  
Feature Reward - Reactor Overload free spins deliver the high value events that drive RTP and the random multiplier creates variety.

# 2. Symbol Set

There are 8 symbols across 3 tiers, plus one scatter. These are distributed like 3 of Low tier symbols, 3 of High tier symbols, 1 Wild symbol and 1 Scatter symbol.

## 2.1. PayTable

Multipliers are per payline, applied on total bet. Win pay are left to right from reel 1

| Symbol               | 3 of a kind | 2 of a kind | Notes |
|----------------------|-------------|-------------|-------|
| Symbol A [Low]       | 5x          | 0.5x        |       |
| Symbol K [Low]       | 10x         | 1x          |       |
| Symbol Q [Low}       | 20x         | 2x          |       |
| Dilution Coil [High] | 40x         | 4x          |       |
| Cryostat Ring [High] | 75x         | 8x          |       |
| Qubit Cluster [High] | 200x        | 20x         |       |

|                       |      |     |                                   |
|-----------------------|------|-----|-----------------------------------|
| Quantum Core [Wild]   | 500x | 50x | Substitute all except Energy Cell |
| Energy Cell [Scatter] | -    | -   | Charges Reactor Overload Meter    |

The Wild tier will substitute all except Energy Cell Scatter. Minimum 2 of a kind symbol is required to appear on reel 1 and 2 for valid payline.

## 2.2. Symbol Description

- 2.2.1. Quantum Core (Wild) - Gold spherical outer rings with blue plasma core. Encapsulating the blue plasma energy.
- 2.2.2. Qubit Cluster (High) - Interconnected 5 golden spheres on a black chip. It resembles a 5 qubit processor tile.
- 2.2.3. Cryostat Ring (High) - A stack of 3 dilution refrigerator rings made out of gold with thermal pads around it.
- 2.2.4. Dilution Coil (High) - A coil made of superconductor, gold braided with blue glow of superconductor.
- 2.2.5. Energy Cell (Scatter) - A transparent blue cylinder with gold casing on either side with glowing blue energy inside it.
- 2.2.6. Symbol A, K, Q (Low) - Red, Blue and Green alphabets with Gold stroke.

## 3. Mathematics

Figures in this section are design targets. The final probability should be determined by testing and confirming before release.

### 3.1. RTP Targets

| Events                                                | RTP Contribution | Notes                  |
|-------------------------------------------------------|------------------|------------------------|
| Base game payline wins                                | 75.50%           | For 5 paylines         |
| Feature - Reactor Overload (free spins + multipliers) | 20%              | Primary Feature        |
| Total                                                 | 95.50%           | Tolerance - $\pm 0.10$ |

### 3.2. Volatility & Frequency

The game targets high volatility. A 25% hit frequency is because the large majority of hits are micro wins of 0.5x to 3x bet these extend the session length.

The Reactor Overload Feature triggers roughly every 100 spins, this is where the mass RTP is. The player can spin 80 times and still get no feature or can hit 2 features close together. This 20% RTP feature is intentional.

### 3.3. Win Distribution

| Wins Types  | Approx. % of spins | Win range      | Source                     |
|-------------|--------------------|----------------|----------------------------|
| No Win      | ~75%               | 0x             |                            |
| Micro Win   | ~20%               | 0.5x - 3x      | Low Symbol Wins            |
| Small Win   | ~4%                | 4x - 30x       | Symbol Combos              |
| Feature Win | ~1%                | 30x - 250x avg | Reactor Overload Free Spin |

## 4. Reactor Overload Feature

The Reactor Overload is the primary feature. It is triggered when the Reactor Overload Meter is filled, which is charged by landing Energy Cells Scatter Symbols. The Feature rewards free spin with a random multiplier on every spin.

### 4.1. Reactor Overload Meter

Reactor Overload Meter is a 10 segmented bar displayed at the top of the game panel. It is styled as a horizontal reactor charge same as Energy Cell.

**Capacity** - 10 Segmented bar, triggers when the meter reaches 10/10.

**Energy Cells** - Each Energy Cell visible after a spin adds +1 charge and a maximum of +3 charge can be obtained per spin.

**Meter Progression** - The meter value persists for the duration of the session. It resets to zero only after the Reactor Overload Feature is completed.

**Cross Sessions** - The meter does not continue across sessions. It reloads to zero when the session ends.

**During the Feature** - During the free spin, the meter is inactive. Energy Cells are removed from the spin.

**Overcharge** - The meter caps at the 10 and the feature triggers immediately. Eg - If the meter is on 9 and 2 Energy cells spawns as soon as the first cell is counted the feature will trigger.

### 4.2. Free Spin Rules

**Reward** - 8 Free spins, fixed count and No retrigger.

**Reels** - The 3x3 grids backend swaps to a modified reel set that replaces Energy Cells position with additional Quantum Core (Wild) symbols.

**Quantum Core Multiplier** - When the spin starts the backend independently draws one multiplier value for that spin. The draw applies to the total win on any payline that contains at least one Quantum Core (Wild). Multiple Wilds on the same payline do not stack the multiplier in this case a single value is used.

**Meter** - Meter is disabled during the feature and is reset to zero.

### 4.3. Quantum Core Multiplier Table

One Value is drawn per free spin. The value is displayed to the player once the backend has returned a spin result.

| Multiplier | Probability |
|------------|-------------|
| 2x         | 40%         |
| 3x         | 25%         |
| 5x         | 18%         |
| 8x         | 10%         |
| 10x        | 5%          |
| 15x        | 2%          |

### 4.4. Feature Presentation

This is the presentation on the front end. No logic is performed on the client side.



## 4.5. Player Psychology

Reactor Overload is designed around one of the most successful bonus structures in slot games.

**Anticipation and Reward** - Player Learns quickly that landing the required symbol activates Reactor Overload. This creates a clear objective base gameplay. When a symbol that triggers feature meter increase appears on the reels it naturally increases the anticipation and this is the main reason for increase in player engagement.

**Familiarity** - As per research in the industry it shows that players prefer bonus mechanics that are easy to understand and support immediate reward. Free spin requires very little explanation and is considered a valuable reward. With this player can focus on the excitement of the feature rather than learning it.

**Variable Multiplier and Near miss moments** - At the start the Feature multiplier is randomly selected this changes from spin to spin. This variable carries anticipation and keeps players engaged. Near miss moments create a memorable peak experience. High multiplier value combined with paylines can produce substantial payout.

## 5. Jackpot

Quantum Jackpot includes 2 tiers. Both tiers are rewarded via random chance draw on non winning base game spins. All jackpot logic is server authoritative.

2 Jackpot is because they are sufficiently rewarding and easier for players to understand and easier to build.

### 5.1. Jackpot Tiers

| Tiers | Name    | Prize          | Approx Avg Frequency |
|-------|---------|----------------|----------------------|
| 1     | Fusion  | 100x total bet | ~1 in 500 spins      |
| 2     | Quantum | 500x total bet | ~1 in 2500 spins     |

### 5.2. Award Rules

- Jackpots are evaluated on non winning base game spins only. Free spins are excluded.
- When a zero win is confirmed, the back end performs an independent random draw for each tier.
- The draw probability for each tier is a configurable server side parameter. The front end never stores or calculates the jackpot.

- The jackpot wins are credited to the player balance as a separate transaction, apart from the spin base win which is zero.
- The price amounts are fixed multipliers of the player's current total bet with no progressive pool.

### 5.3. Display

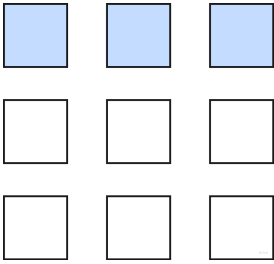
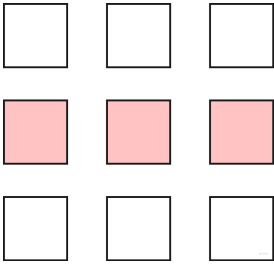
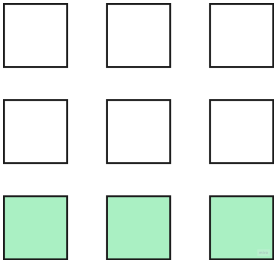
- The Jackpot price values will be displayed on a strip at the bottom of the game panel.
- ON winning the Reward a full screen celebration overlays fires with tier specific styling. Gold - Fusion, White and Blue particles for Quantum.

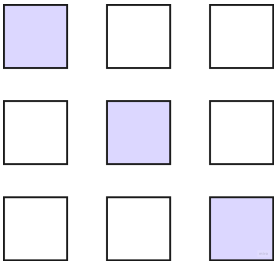
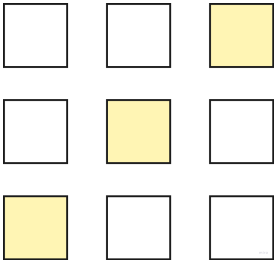
## 6. Paylines

There will be 5 fixed paylines on every spin. Win pay is left to right from reel 1.

A winning combination requires a matching symbol or Wild substitution on consecutive reels starting from reel 1.

The 5 lines cover all the three rows and both cross diagonal. This covers a full grid with a clean and recognisable set.

| # | Row Pattern (R1-R2-R3)   | Position Description | Shape                                                                                 |
|---|--------------------------|----------------------|---------------------------------------------------------------------------------------|
| 1 | Top - Top - Top          | Top Horizontal       |  |
| 2 | Middle - Middle - Middle | Center Horizontal    |  |
| 3 | Bottom - Bottom - Bottom | Bottom Horizontal    |  |

|   |                       |                                   |                                                                                     |
|---|-----------------------|-----------------------------------|-------------------------------------------------------------------------------------|
| 4 | Top - Middle - Bottom | Diagonal top left to bottom right |  |
| 5 | Bottom - Middle - Top | Diagonal bottom left to top right |  |

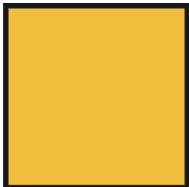
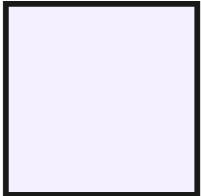
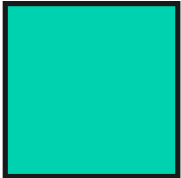
## 7. Art Direction

### 7.1. Visual reference

The Games theme directly draws from real Quantum Computer hardware. The Key visual revolves around Gold, White and Blue Plasma Energy. The dilution refrigerator in the IBM Quantum system communicates “advanced technology” without explanation. Those who know quantum computers will recognize it and those who do not will still find it spectacular.

### 7.2. Colour Palette

| Role       | Colour & Hex                                                                                   | Name            | Used For                     |
|------------|------------------------------------------------------------------------------------------------|-----------------|------------------------------|
| Background | <br>#0A1628 | Deep Space Navy | Game panel, reel background. |

|                |                                                                                                            |                |                                     |
|----------------|------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------|
| Primary Metal  | <div><br/>#B8962E</div>   | Quantum Gold   | Rings, Frame, premium symbols       |
| Accent Metal   | <div><br/>#F0C040</div>   | Bright Gold    | Win highlights, UI Texts            |
| Energy         | <div><br/>#1A78C2</div>  | Electric Blue  | Meter fills, glows, energy arcs     |
| Overload State | <div><br/>#F5F0FF</div> | White Gold     | Reel frame during free spin         |
| Thermal Accent | <div><br/>#00D4B0</div> | Cryogenic Teal | Symbol thermal detail, H2 ring pads |

### 7.3. Structural Elements

- **Background** - The BG behind the reel will be of Deep Space Navy color with golden wires dangling (the wires will not be connected to anything). The dangling wires will be with 40% opacity and blurred. The wires will be in different layers giving the BG a parallax effect.
- **Cables** - 2 Cables will connect the reel to the Reactor Overload meter. The cables will be braided. They will be animated with sway on loop. On a big win the Blue energy will create blue sparks.
- **Reel Frame (Base)** - The frame will be made of thick Gold and will have teal padding similar to the real life quantum computer.
- **Reel Frame (Overload)** - The frame after receiving energy from the overload will turn into iridescent finish with Blue energy leaking out like sparks, arcs.

## 8. Sound Design

### 8.1. Direction

The SFX will be of cinematic Sci Fi electronic ambient. The background music will be deep bass machinery with melodic pads and high tones instrumental for adrenaline. Wins Feature trigger will cut ambient music.

### 8.2. Music

- **Base Game** - 80~90 seconds seamlessly looped ambient audio. Deep bass, melodic mid range pads and high tones.
- **Reactor Overload** - Intense layered, Faster pulse, aggressive bass, new synth. This cross fades into ambient for feature.

### 8.3. Sound Effects

| Event                             | Description                                    |
|-----------------------------------|------------------------------------------------|
| Spin Initiated                    | Electromagnetic Whirr                          |
| Reel Stop (x3, Pitch Low to High) | Magnetic clunk, distance pitch                 |
| Small Win (0.5x - 4x)             | Electronic Chime                               |
| Medium Win (5x - 30x)             | Ascending tone Cascade                         |
| Big win (31x - 99x)               | Orchestral stab and Electronic Swell           |
| Epic Win (100x +)                 | Full climax, 3 s                               |
| Win counter tick                  | Rapid electronic tick while balance increments |
| Energy Cell land                  | Magnetic clunk + plasma hiss                   |

|                          |                                                     |
|--------------------------|-----------------------------------------------------|
| Meter segment fills      | Rising electronic ping, pitch increases per segment |
| Meter 8–9/10 (warning)   | Low pulse + tension drone                           |
| Reactor Overload trigger | Deep bass boom + energy crackle                     |
| Free spin (each spin)    | Higher-pitch reel spin with electrical arcing       |
| Multiplier draw          | Electronic rattle slows to a lock tone              |
| Wild lands in free spin  | Quantum resonance hum + gold shimmer                |
| Overload exit            | Power-down hum, systems offline                     |
| Jackpot win (Fusion)     | Gold orchestral swell                               |
| Jackpot win (Quantum)    | Full orchestral + cinematic VO: 'QUANTUM JACKPOT!'  |

## 9. Front End Responsibilities

The front end is only for presentation. It renders the result provided by the server and handles player input. It does not perform any game logic, win calculation or RTP computing.

### 9.1. What the Front End Does

- Sends spin request to the server (bet amount, session token)
- Receives the spin result payload and animates the outcome. Reel stop position, winning paylines, meter update.
- Manages all symbol animation and feature entry exit effects.
- Display the Reactor Overload meter state, Jackpot prize value, Balance and bet control.
- Handles player input like Spin, Auto Spin, Bet adjust, Paytable, Settings.
- Manages audio payback in response to game events.

### 9.2. What the Front End Does Not Do

- No win calculation. All payline evaluation happens on the server.
- No RNG. The server determines all reel outcomes.
- No RTP tracking or jackpot probability Computing.
- No game state in stored client side beyond display preferences.

### 9.3. State Machine

The client manages a simple finite state machine to control animation sequencing and prevent conflicting requests.

- **IDLE** - Awaiting Player Input
- **SPINNING** - Spin request sent, reels in motion.
- **WIN PRESENTATION** - Animating Win highlights and computer.
- **FEATURE ENTRY** - Reactor Overload animation sequence playing.
- **FREE SPIN** - Active spin in progress.
- **FEATURE EXIT** - Post feature animation and result display.
- **JACKPOT** - Jackpot celebration playing.
- No new spin request can be sent while the client is in any state other than IDLE.

## 9.4. Development Estimate

| Task                                               | Days | Notes                      |
|----------------------------------------------------|------|----------------------------|
| Project setup, reel engine, basic spin loop        | 5    |                            |
| Symbol art integration + win animation             | 5    | 8 Symbol                   |
| UI - bet, balance, spin button, autoplay, paytable | 4    |                            |
| Reactor Overload Meter + Energy Cell               | 3    |                            |
| Reactor Overload entry, free spin, exit sequence   | 6    | Includes Multiplier reveal |
| Jackpot display + celebration (2 tiers)            | 2    |                            |
| Audio integration                                  | 2    |                            |
| API integration + error handling                   | 3    |                            |
| QA and cross browser testing                       | 5    | Desktop + Mobile           |
| Total                                              | 35   | ~7 weeks                   |

## 10. Back End Responsibilities

All Game logic resides on the server. The following items must never be delegated to the client.

### 10.1. Server Authority

- RNG - PRNG output is never exposed to the client.
- Reel strip management and Symbol position.
- Payline evaluation and win calculation across all 5 lines.
- Wild Substitution logic.
- Reactor Overload Meter state and trigger decision.
- Free spin reel set swap (Energy Cell position replaced by additional Wild position)
- Quantum Core Multiplier draw per free spin.

- Jackpot draws evaluation and reward.
- Max win cap enforcement
- Balance management and transaction record.

## 10.2. Reel Sets

2 reel sets are required. The Server selects the active set based on game state.

- Base reel set - Used during normal play. Contains all symbols including Energy Cells.
- Free spin reel set - Used during Reactor Overload free spin. Energy Cells are replaced with additional Quantum Core Multiplier rest symbols that remain the same. The front end only receives the resulting symbol.

## 10.3. Spin response Payload

- **reelResult** - 3x3 grid of symbols.
- **wins** - Array of paying win objects (Payline number, symbol, count, amount)
- **totalWin** - Total win amount for the spin.
- **meterLevel** - Reactor Overload value after the spin. Not returning during free spins.
- **featureTrigger** - Boolean. True if the Reactor Overload triggers this spin.
- **freeSpinMultiplier** - Integer. Multiplier value draw for this spin.
- **freeSpinRemaining** - Integer. Only present during active features.
- **jackpotWin** - Null or object with tier.
- **Balance** - Player balance after the spin is settled.

## 10.4. Development Estimate

| Task                                        | Days | Notes                     |
|---------------------------------------------|------|---------------------------|
| Certifies PRNG integration                  | 3    |                           |
| Reel strip design + RTP modelling           | 5    | Target 95.50% $\pm$ 0.10% |
| Payline engine + Wild substitution          | 3    |                           |
| Energy Cell + Meter + feature trigger       | 3    |                           |
| Free spin mode + multiplier draw            | 3    |                           |
| Jackpot draw + award + cap enforcement      | 3    |                           |
| Session, balance, transaction ledger        | 3    |                           |
| API endpoints + RTP simulations (10M Spins) | 4    |                           |
| Integration testing with front end          | 4    |                           |
| Total                                       | 31   | ~6 Weeks                  |

Running Front End and Back End in parallel with shared QA brings the total project duration to approx 10 weeks.

## 11. Responsible Gambling

- **Reality Check** - A session time reminder, displays elapsed time and net session result.
- **Auto spin control** - Max 100 auto spin pause if single win exceeds a threshold or balance drops.
- **Spin history** - Last 20 spins are accessible in game, showing bet, win, and running balance.
- **Meter** - The Reactor Overload meter resets on session end. This avoids rewarding compulsive re engagement.
- All responsible gambling is operated via RGS layers.

## 12. Design Decisions

| Decision                                         | Chosen Approach                                                                | Rationale                                                                                                         |
|--------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Does the Overload Meter persist across sessions? | Does the Overload Meter persist across sessions?<br>No. Resets on session end. | Avoids regulatory compulsion concerns. Simpler server state.                                                      |
| Can the feature retrigger?                       | No. 8 spins fixed, no retrigger.                                               | Keeps feature duration predictable. The multiplier table provides sufficient variance.                            |
| Fixed or random free spin count?                 | Fixed at 8.                                                                    | Simpler to model mathematically. Random count adds complexity without clear player benefit.                       |
| How many jackpot tiers?                          | Two: Fusion (100×) and Quantum (500×).                                         | Fewer tiers concentrate prize mass and keep the jackpot strip readable. Four tiers felt excessive for a 3×3 game. |
| Local or networked jackpot?                      | Local (per-operator). Fixed prize amounts at current bet.                      | Simpler to implement, certify, and explain to players. No shared pool infrastructure required.                    |
| Max win enforcement method?                      | Hard cap on the server at 2,500× total bet.                                    | Cleanest compliance approach. Cap is applied before win is credited.                                              |